

Notes: Strebulaev (JF, 2007)

Do Tests of Capital Structure Theory Mean What They Say?

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1 Big picture

- **Question.** Do standard empirical tests of capital structure theory really test what they claim to test?
- **Core idea.** Most empirical tests compare cross-sectional leverage ratios with static or refinancing-point predictions of capital structure models. But with adjustment costs, firms refinance only occasionally, so most observed firms are not at their refinancing point when the econometrician observes them.
- **Main problem.** Actual leverage is a state variable shaped by the whole history of past shocks and past financing decisions. It can be far from the leverage a firm would choose if it refinanced today.
- **Method.** The paper builds a calibrated dynamic trade-off model with infrequent adjustment, simulates many artificial economies, and runs standard empirical capital structure tests on the simulated panel data.
- **Main finding.** Standard cross-sectional tests on model-generated data can reproduce several well-known empirical patterns that are often interpreted as evidence against the trade-off theory.
- **Most important example.** At refinancing points, more profitable firms choose more debt, as trade-off theory predicts. But in the simulated dynamic cross section, leverage and profitability are negatively related, matching the empirical literature.
- **Economic mechanism.** A profitability shock raises firm value immediately, mechanically lowering leverage for firms that do not refinance. Since most firms do not refinance in a given period, this mechanical effect dominates the positive target-leverage effect.
- **Broader implication.** Dynamic capital structure models should be judged by their predictions in true dynamics, not merely by comparative statics at adjustment dates.
- **Takeaway.** Some empirical “failures” of the trade-off model may be failures of the test design rather than failures of the model.

2 Incremental contribution of the paper

- **From date-zero theory to dynamic empirics.** Earlier dynamic trade-off models characterize optimal refinancing policies. Strebulaev asks what those policies imply for the cross section observed by empirical researchers.
- **A simulation-based testing framework.** The paper is among the first to use a calibrated dynamic capital structure model to generate artificial Compustat-like data and then run the same regressions used in empirical capital structure papers.
- **Reinterpretation of leverage-profitability evidence.** The paper shows that a negative leverage-profitability relation can emerge from a dynamic trade-off model, even though the same model predicts a positive relation at refinancing points.
- **Reinterpretation of Welch’s stock-return result.** Welch (2004) finds that market leverage is largely explained by past stock returns, suggesting that firms passively allow stock prices

to move leverage. Strebulaev shows that a model with active but infrequent adjustment can generate almost the same pattern.

- **Distinguishing active and passive firms.** The paper demonstrates that regressions on active refinancers look much more like model predictions, while passive firms drive many puzzling cross-sectional results.
- **Quantitative bridge between theory and evidence.** The model is calibrated to match broad features of U.S. firms, including leverage, asset volatility, tax rates, restructuring costs, refinancing costs, and equity issuance costs.
- **Methodological warning.** The paper argues that the null hypothesis in empirical capital structure tests must come from the model's dynamic cross section, not from static comparative statics.

3 Research question and conceptual problem

3.1 The empirical regularities motivating the paper

The paper begins from a set of empirical findings that had become central in the capital structure literature:

- leverage is related to profitability, market-to-book, size, and other firm characteristics;
- changes in market leverage are heavily explained by changes in equity value;
- past book-to-market ratios predict current capital structure;
- firms appear conservative in using debt;
- firms seem to move slowly toward target leverage;
- highly levered and low-levered firms react differently to shocks.

Interpretation: These facts are often compared with static predictions of trade-off, pecking-order, and market-timing theories. Strebulaev's point is that this comparison may be inappropriate if observed leverage is far from target most of the time.

3.2 Why refinancing-point predictions can be misleading

In a frictionless model, firms continuously adjust capital structure, so observed leverage is always close to optimal leverage. In a model with adjustment costs, a firm optimally waits until its cash-flow or value process reaches a refinancing boundary. Therefore:

- refinancing happens discretely;
- firms are observed at different points in their refinancing cycles;
- actual leverage reflects past shocks, not only current target leverage;
- the cross section is a mixture of firms that recently refinanced and firms that have not adjusted for years.

Economic message: Infrequent adjustment breaks the link between the theoretical object “optimal leverage at adjustment” and the empirical object “observed leverage in a random cross section.”

3.3 Central empirical warning

The paper can be summarized as a warning:

A theory should not be tested only by comparing observed leverage with what the theory predicts at refinancing points. The correct comparison is between observed leverage and the cross section generated by the theory in dynamics.

4 Model environment

4.1 All-equity benchmark

The firm owns an infinitely lived project that produces a stream of net payouts. The state variable is total net payout to claimholders, denoted δ_t . Under the pricing measure,

$$\frac{d\delta_t}{\delta_t} = \mu dt + \sigma dZ_t, \quad (1)$$

where μ is the risk-neutral drift and σ is the instantaneous volatility of net cash flow.

For an all-equity firm, shareholder value is

$$E(\delta_0) = (1 - \tau) \frac{\delta_0}{r - \mu}, \quad (2)$$

where $\tau = 1 - (1 - \tau_c)(1 - \tau_d)$ captures corporate and dividend taxation.

Interpretation: This unlevered value provides the baseline against which debt tax benefits, bankruptcy costs, and adjustment costs are evaluated.

4.2 Levered firm and financial policy

The levered firm issues perpetual callable debt. Debt pays a continuous coupon c and can be called at face value. Equity holders choose a dynamic policy that maximizes current equity value, taking into account how debt holders price their claims.

Key ingredients are:

- **Tax benefits of debt.** Interest payments generate corporate tax benefits, but these benefits are partly offset by personal taxation.
- **Default costs.** If the firm defaults, a fraction α of asset value is lost.
- **Adjustment costs.** Issuing or retiring debt involves proportional cost q_{RC} .
- **Liquidity crisis.** When cash flow is too low to cover interest, the firm enters financial distress.
- **Asset sales.** A distressed firm may sell assets, keeping fraction k and using proceeds to reduce debt.

- **Equity issuance.** If distress continues, equity holders may issue equity at cost q_E to cover debt payments.
- **Refinancing boundaries.** If firm value or cash flow rises sufficiently, equity holders refinance upward by calling old debt and issuing a larger amount of debt.

4.3 Key boundaries

The model has several endogenous thresholds:

- δ_U : the upper refinancing boundary for healthy firms;
- δ_{LU} : the upper refinancing boundary after a liquidity crisis;
- δ_L : the liquidity boundary, triggered when cash flow cannot cover coupon payments;
- δ_B : the default boundary;
- c : the coupon chosen at refinancing.

Interpretation: The firm's policy is not to continuously rebalance leverage. Instead, it waits until a boundary is crossed and then takes a discrete action.

4.4 Refinancing-cycle logic

A refinancing cycle begins when a firm chooses a coupon and debt level. During the cycle:

- if cash flows rise enough, the firm refinances upward;
- if cash flows fall below the liquidity boundary, the firm sells assets and reduces debt;
- if the firm recovers after distress, it can refinance again;
- if the firm continues to deteriorate, it defaults and debt holders receive the firm.

Economic message: The model is a dynamic trade-off theory with realistic inaction regions. Firms are always following an optimal policy, but optimal policy often means doing nothing.

5 Comparative statics at refinancing points

At refinancing points, the model behaves like a standard dynamic trade-off model. The main comparative statics are:

- higher corporate or dividend tax rates increase the value of debt tax shields and tend to raise optimal leverage;
- higher personal taxes on interest reduce the attractiveness of debt;
- higher cash-flow volatility lowers leverage because distress and bankruptcy become more likely;
- higher bankruptcy costs lower leverage;
- higher asset sale costs and equity issuance costs lower leverage;

- adjustment costs have a nonmonotonic effect on leverage: modest costs can lead firms to borrow more and wait longer, but high costs make debt less attractive overall.

Interpretation: If one looked only at refinancing points, the paper would appear to deliver standard trade-off predictions. The surprises appear only after the model is simulated through time.

6 Simulated dynamic economy

6.1 Why simulation is needed

The model is nonlinear, and the cross section is shaped by firms' histories. The impact of a shock depends on where the firm is in its refinancing cycle. Therefore, simple analytic comparative statics are not enough to characterize the observed dynamic cross section.

6.2 Systematic and idiosyncratic shocks

For simulations, the cash-flow process is decomposed into an idiosyncratic component and a systematic component:

$$\frac{d\delta_t}{\delta_t} = \mu dt + \sigma_I dZ_I + \beta \sigma_S dZ_S. \quad (3)$$

Here:

- $\sigma_I dZ_I$ is the firm-specific shock;
- $\beta \sigma_S dZ_S$ is the systematic economy-wide or industry-wide shock;
- β governs exposure to the systematic shock;
- total volatility is $\sigma = (\sigma_I^2 + \beta^2 \sigma_S^2)^{1/2}$.

Interpretation: Systematic shocks are important because cross-sectional regression coefficients depend on the realization of common shocks.

6.3 Simulation design

The benchmark simulation proceeds as follows:

- simulate 3,000 firms;
- simulate 300 quarters, or 75 years;
- drop the first 148 quarters to reduce dependence on initial conditions;
- retain 152 quarters, or 38 years, for analysis;
- repeat the procedure for 1,000 simulated economies;
- run standard empirical capital structure regressions in each simulated economy;
- average regression coefficients and test statistics across economies.

Interpretation: The paper does not simply solve for an optimal policy. It asks what an empirical researcher would find if the world were generated by that optimal policy.

6.4 Calibration

The calibration is designed to resemble firms used in empirical studies. Key choices include:

- initial firm value and book assets scaled to 100;
- beta distribution estimated from CRSP market-model regressions;
- asset return volatility calibrated to Schaefer and Strebulaev's estimates;
- refinancing costs chosen to be small but positive;
- restructuring, asset-sale, and equity-issuance costs drawn from plausible empirical ranges;
- corporate tax rate set to 35%;
- personal tax rates based on Graham's estimates;
- after-tax risk-free rate set to 5%;
- asset risk premium set to 6.5%.

Interpretation: The model is stylized but quantitatively calibrated. It is not meant to fit every feature of the data, but to test whether a plausible dynamic trade-off model can generate the main empirical patterns.

7 Variables used in simulated empirical tests

7.1 Market leverage

Market leverage is

$$ML_t = \frac{D(\delta_t)}{F(\delta_t)}, \quad (4)$$

where $D(\delta_t)$ is market debt value and $F(\delta_t)$ is total firm value.

7.2 Quasi-market leverage

Because empirical work often uses book debt and market equity, the paper defines quasi-market leverage as

$$QML_t = \frac{D_0(\delta_t)}{D_0(\delta_t) + F(\delta_t) - D(\delta_t)}. \quad (5)$$

This is close to common empirical measures of debt-to-capital.

7.3 Profitability

Profitability is

$$\pi_t = \frac{\delta_t + \Delta A_t}{A_{t-1}}, \quad (6)$$

where A_t is book assets. This corresponds to earnings before interest and taxes over book assets.

7.4 Implied debt ratio

Following Welch (2004), the implied debt ratio asks what leverage would be if firms had not issued or retired securities between $t - k$ and t . This variable isolates the mechanical effect of equity returns on leverage.

8 Main empirical exercises

8.1 Descriptive leverage distribution

The first comparison is between leverage at refinancing points and leverage observed in the dynamic cross section.

Key result: Dynamic leverage is higher and more dispersed than refinancing-point leverage. Average market leverage is about 0.26 at refinancing points but about 0.36 in dynamics.

Interpretation: Observed leverage ratios in dynamic data are not simply noisy target leverage ratios. They reflect firms' positions in their refinancing cycles.

8.2 Cross-sectional leverage regressions

The paper runs regressions similar to Bradley, Jarrell, and Kim (1984), Rajan and Zingales (1995), and Fama and French (2002):

$$QML^P = d_0 + d_1\pi^P + d'_x x + \varepsilon^P, \quad (7)$$

where x includes firm technology parameters such as volatility and costs, and P indexes the empirical procedure.

Main contrast:

- at refinancing points, the coefficient on profitability is strongly positive;
- in simulated dynamic cross sections, the coefficient on profitability is negative;
- the Fama-French-style coefficient is about -0.78 , similar to empirical estimates.

Economic message: A negative leverage-profitability relation does not by itself reject the dynamic trade-off theory.

8.3 Welch-style leverage and stock return tests

The Welch-style regression is

$$QML_t = f_0 + f_1 IDR_{t-k,t} + f_2 QML_{t-k} + \varepsilon. \quad (8)$$

If f_1 is close to 1, observed leverage behaves as if firms passively let stock returns move leverage.

Key result: The model generates coefficients very close to Welch's estimates. For a one-year horizon, the simulated IDR coefficient is about 1.022, while Welch's empirical estimate is about 1.014.

Interpretation: A firm can be active over long refinancing cycles but look passive in short-horizon regressions. The apparent passivity comes from infrequent adjustment, not from managerial indifference.

8.4 Mean reversion of leverage

The paper estimates partial-adjustment models of leverage changes:

$$QML_t - QML_{t-1} = h_0 + h_1 TQML_{t-1} + h_2 QML_{t-1} + h_3 X_{t-1} + \varepsilon. \quad (9)$$

Key result: Leverage mean reverts at about 17% per year, consistent with empirical findings often described as slow adjustment.

Interpretation: The paper calls firms “crouching tigers”: they usually do nothing, but when they act, they adjust by a large amount.

8.5 Active versus passive firms

The paper separates firms that refinanced in the last year from firms that did not.

- For active firms, cross-sectional regressions have much higher explanatory power.
- The profitability coefficient is nearly zero for active firms.
- The negative profitability relation is concentrated among passive firms.

Economic message: Empirical tests should condition on the firm’s refinancing state. Mixing active and passive firms can generate misleading slopes.

9 Identification and empirical design

This paper is not a causal treatment-effect study. Its empirical design is a model-based diagnostic exercise.

9.1 What is being identified?

The exercise identifies whether a dynamic trade-off model, when viewed through the same econometric lens as empirical researchers use, can reproduce empirical regularities.

9.2 What makes the exercise informative?

- The model parameters are calibrated to empirical moments and plausible cost estimates.
- The simulated data have the same kind of variables used in the empirical literature: leverage, profitability, stock returns, volatility, and lagged leverage.
- The regressions are deliberately close to established empirical procedures.
- Because the data-generating process is known, the researcher can see whether standard interpretations are valid.

9.3 Main identifying logic

If a standard empirical test rejects a model when run on data generated by that same model, then the test’s interpretation is unreliable.

Interpretation: The paper’s main contribution is diagnostic. It shows that common empirical tests may be miscalibrated against the wrong theoretical benchmark.

10 Core economic intuition

10.1 The leverage-profitability reversal

At the refinancing point, more profitable firms borrow more because expected distress costs are lower and tax shields are more valuable. But after refinancing, most firms do not adjust. A positive profitability shock raises equity value and total firm value immediately. Debt is sticky, so leverage falls.

- **Refinancing-point effect:** profitability $\uparrow \Rightarrow$ target leverage $\uparrow$.
- **Within-cycle effect:** profitability $\uparrow \Rightarrow$ firm value $\uparrow \Rightarrow$ observed leverage $\downarrow$.
- **Cross-sectional dominance:** since passive firms dominate the sample, the within-cycle effect dominates.

10.2 The stock-return result

When stock prices rise, market leverage mechanically falls. If the firm does not refinance immediately, the stock-return effect persists. Infrequent adjustment makes the implied debt ratio explain leverage almost one-for-one over short horizons.

Economic message: Welch’s finding that firms do not offset stock returns is consistent with optimal inaction under adjustment costs.

10.3 The low leverage puzzle

Dynamic models are often judged by leverage at refinancing points. Strebulaev emphasizes that empirical leverage should be compared with dynamic average leverage, not refinancing-point leverage.

Economic message: The “low leverage puzzle” may be partly about comparing the wrong empirical object to the wrong theoretical object.

11 Limitations of the model

The paper is explicit about what the model cannot explain.

- **No endogenous investment.** Investment policy is exogenous, so the model cannot explain financing deficits or the interaction between investment and financing.
- **No equity issuance in good states.** Equity is issued only in distress, so the model cannot explain all patterns involving market timing or high market-to-book ratios.
- **Proportional rather than fixed debt adjustment costs.** The model cannot directly explain the size-leverage relation.

- **No cash policy.** Firms do not choose optimal cash holdings jointly with debt.
- **Perfect correlation of cash flow and firm value.** This creates unrealistically high correlations between some variables, such as profitability and market-to-book.

Interpretation: The paper is not claiming that its model explains all capital structure evidence. It shows that even a stylized dynamic trade-off model can overturn conventional interpretations of key tests.

12 Tables and figures

12.1 Figure 1 and Table I: model paths and refinancing-point comparative statics

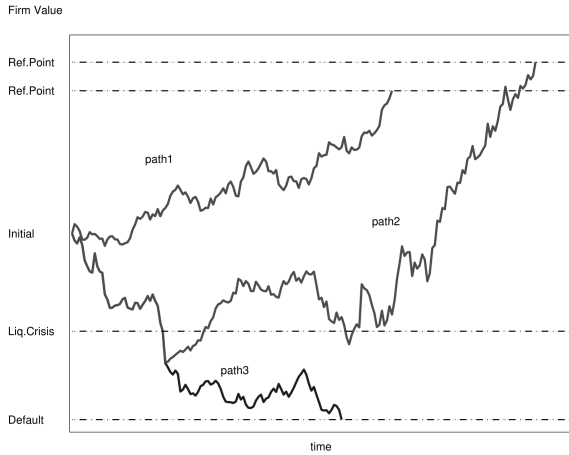


Figure 1. Possible paths of firm value. The figure shows possible model scenarios. Path 1 depicts a successful firm that refinances when firm value increases substantially. Paths 2 and 3 show firms that face a liquidity crisis. After selling assets and issuing equity the firm in Path 2 recovers and refinances when it reaches the upper restructuring threshold. The firm in Path 3 does not recover and equity holders decide to default when firm value is sufficiently low.

Table I
Comparative Statics of Financial Variables at the Refinancing Point

The table gives the comparative statics at the refinancing point of the following variables: the optimal market leverage ratio (ML), bankruptcy boundary (δ_B), restructuring boundaries (δ_U and δ_{UL}), total firm value ($F(\delta_0)$), coupon rate (c), and liquidity barrier (δ_L). The corporate tax rate is τ_c , τ_d is the dividend tax rate, τ_i is the interest income tax rate, r is the pre-tax risk-free interest rate, σ is the volatility of the firm's cash flow, α is the fraction of asset value lost in bankruptcy, q_{RC} is the adjustment cost, q_A is the cost of selling assets in a liquidity crisis, q_E is the cost of equity issuance, and k is the fraction of asset value that remains after an asset sale.

Variable	Shape	Sign of Change in Variable for an Increase in:								
		τ_c, τ_d	τ_i	r	σ	α	q_{RC}	q_A	q_E	k
ML	Invariant to δ	>0	<0	<0	<0	<0	>0, q_{RC} small <0, q_{RC} large	<0	<0	>0
δ_B	Linear in δ	>0	<0	<0	<0	<0	>0, q_{RC} small <0, q_{RC} large	<0	<0	>0
δ_U, δ_{UL}	Linear in δ	<0	>0	<0	>0	<0	>0	>0	>0	<0
$F(\delta_0)$	Linear in δ	<0	<0	>0	<0	<0	<0	<0	<0	>0
c, δ_L	Linear in δ	>0	<0	>0	<0	<0	>0, q_{RC} small <0, q_{RC} large	<0	<0	>0

Figure 1. Possible paths of firm value

Table I. Comparative statics at the refinancing point

Read: Figure 1 shows three representative firm histories. A successful firm reaches an upper refinancing point and re-levers. A distressed firm may enter a liquidity crisis, sell assets, recover, and refinance. Another distressed firm may continue to deteriorate and default. Table I summarizes how refinancing-point variables respond to model parameters.

Economic message: The model produces standard trade-off predictions at refinancing points. The puzzle is not in the model's target leverage; it is in how target leverage translates into observed leverage in dynamics.

12.2 Table II and Table III: calibration and dynamic leverage distribution

Table II
Parameter Values for Benchmark Simulations

Listed are the values and sampling distributions chosen for all parameters required to simulate the benchmark case of the model. V_0 is the present value of all future net payouts at time 0, A_0 is the initial book value of firm assets, β is the systematic risk of the firm's assets, σ_E is the volatility of monthly returns on the CRSP value-weighted equity return index, σ_D is the volatility of monthly returns on the CRSP 10-year T-note index over the period 1965 to 2000, σ_{ED} is the covariance between equity and debt returns, L_{av} is average leverage computed from annual COMPUSTAT data for 1965 to 2000, averaging first for each year over firms and then over time, and defined as the ratio of book debt to the sum of book debt and market equity, σ_I is the volatility of idiosyncratic shocks, σ is the instantaneous volatility of the project's net cash flow, q_A is the proportional costs incurred in selling assets, q_{RC} is the proportional adjustment costs of issuing/retiring debt, q_E is the proportional direct costs of external equity financing, α is the proportional restructuring costs, k is the fraction of assets that remains after an asset sale, κ defines the partial loss-offset boundary, g is the growth rate of book assets, a is the shift parameter in the net payout ratio estimation, RP_A is the asset risk premium, τ_c is the loss per dollar of full offset in the case of distress, τ_i is the marginal corporate tax rate, τ_d is the marginal personal tax rate on dividends, τ_l is the marginal personal tax rate on interest income, and r is an instantaneous after-tax riskless rate. $\mathcal{U}$ indicates uniform distribution.

Parameter	Distribution	Mean	Std. Dev.
V_0	Constant	100	
A_0	Constant	100	
β	Empirical	0.993	0.47
σ_E	Constant	0.155	
σ_D	Constant	0.081	
σ_{ED}	Constant	0.023	
L_{av}	Constant	0.314	
σ_I	$a_0 + a_1 \chi^2(n)$	0.22	0.107
	$[a_0, a_1, n] = [0.05, \frac{1}{30}, 5]$		
σ	Empirical	0.255	0.10
q_{RC}	$\mathcal{U}(0.0005, 0.0025) + 0.001s$	0.002	0.0006
q_E	$\mathcal{U}(0.02, 0.06) + 0.02s$	0.05	0.013
α	$\mathcal{U}(0.03, 0.077) + 0.023s$	0.065	0.015
q_A	$\mathcal{U}(0.05, 0.183) + 0.067s$	0.15	0.043
k	$\mathcal{U}(0.6, 1)$	0.8	0.116
κ	$\mathcal{U}(0.7, 0.9)$	0.8	0.058
g	Constant	$\mu + RP_A$	
a	$\mathcal{U}(0.03, 0.04)$	0.035	0.003
RP_A	Constant	0.065	
τ_c	Constant	0.5	
τ_i	Constant	0.35	
τ_d	Constant	0.351	
τ_l	Constant	0.122	
r	Constant	0.05	

Table II. Benchmark simulation parameters

Table III
Descriptive Statistics

The table reports descriptive statistics for the following variables: market leverage (ML), quasi-market leverage (QML), interest coverage ratio (the ratio of net payout, δ , to coupon, c), tax advantage to debt (i.e., the increase in firm value if the firm moves from no-leverage to its optimal leverage ratio, given by the formula $\frac{F(\delta_t) - (1 - \tau)V(\delta_t)}{(1 - \tau)V(\delta_t)}$, where F is firm value, τ is the effective tax rate, V is the value of firm assets, and δ_t is the level of cash flow at time t). Ref. point refers to the case in which all firms are at their refinancing points. All other statistics are given for dynamics. One thousand data sets are generated, each containing 75 years of quarterly data for 3,000 firms. For each data set the statistics are first calculated for each year in the last 35 years of data and then are averaged across years. Finally, they are averaged over data sets. Min and Max give the minimum and maximum of the annual averages over the 1,000 data sets.

	Mean	Percentiles					Std. Dev.	N
		1%	50%	90%	95%	99%		
<i>Market leverage, ML</i>								
Ref. point	0.26	0.04	0.27	0.40	0.43	0.50	0.10	3,000
Average	0.36	0.06	0.34	0.56	0.66	0.87	0.16	3,000
Min	0.30	0.06	0.29	0.46	0.53	0.76	0.13	3,000
Max	0.43	0.07	0.41	0.71	0.80	0.94	0.20	3,000
<i>Quasi-market leverage, QML</i>								
Average	0.37	0.06	0.35	0.59	0.70	0.91	0.17	3,000
Min	0.31	0.06	0.29	0.47	0.56	0.82	0.14	3,000
Max	0.44	0.07	0.42	0.74	0.84	0.96	0.21	3,000
<i>Interest coverage ratio</i>								
Ref. point	3.98	2.01	3.22	5.74	7.80	17.83	3.24	3,000
Average	3.08	0.69	2.64	4.78	6.08	11.26	2.35	3,000
<i>Tax advantage to debt</i>								
Ref. point	0.05	0.02	0.05	0.07	0.07	0.08	0.01	3,000
Average	0.04	0	0.04	0.07	0.08	0.09	0.02	3,000

Table III. Descriptive statistics in dynamics versus refinancing points

Read: Table II lists the parameter distributions used in the benchmark simulation: tax rates, risk-free rate, asset risk premium, volatility, adjustment costs, default costs, equity issuance costs, asset-sale costs, and firm-specific heterogeneity. Table III compares leverage and other variables at refinancing points and in simulated dynamics.

Economic message: The simulation is calibrated to realistic magnitudes. Dynamic leverage is materially different from refinancing-point leverage: average market leverage rises from 0.26 at the refinancing point to 0.36 in dynamics.

12.3 Table IV and Table V: leverage-profitability and stock-return regressions

Table IV
Cross-Sectional Regressions

The table reports the results of cross-sectional regressions on the level of the quasi-market leverage ratio, QML . One thousand data sets are generated, each containing 75 years of quarterly data for 3,000 firms. Coefficients and t -statistics are means over 1,000 simulations. Independent variables are profitability (π), volatility of cash flows (σ), bankruptcy costs (α), asset sale costs (q_A), and restructuring costs (q_{RC}). The Ref. Point column gives the results obtained by running the regression at the refinancing point. The BJK, RZ, and FF columns report the results of regressions that replicate the empirical procedures used, respectively, by Bradley, Jarrel, and Kim (1984), Rajan and Zingales (1995), and Fama and French (2002). Each of the regressions is of the form:

$$QML^P = d_0 + d_1\pi^P + \mathbf{d}'\mathbf{x} + \epsilon^P,$$

where $\mathbf{x}$ are firm technology parameters and $P, P \in \{\text{BJK}, \text{RZ}, \text{FF}\}$, refers to the method. For BJK and FF, $QML^P = QML_{\text{end}}$ and $\pi^P = \pi_{\text{end}-1}$; for RZ, $QML^{RZ} = QML_{\text{end}}$ and $\pi^{RZ} = \frac{1}{4} \sum_{i=\text{end}-4}^{\text{end}-1} \pi_{it}$, where "end" is the last year in each data set. FF uses the Fama-MacBeth (1973) method, with the regressions run over the last 35 years of each data set and then averaged. The last three columns report additional information on the FF regression: the standard deviation of coefficients and t -statistics, and the 10th and 90th percentile values of these coefficients across simulations. BJK and RZ regression standard errors are standard. FF standard errors are Fama-MacBeth (1973) with the Newey-West correction.

	Ref. Point (1)	BJK (2)	RZ (3)	FF			
				Coeff. (4)	Std. Dev (5)	10% (6)	90% (7)
Constant	0.24 (22.29)	0.61 (29.36)	0.58 (28.06)	0.62 (34.02)	0.06 (21.03)	0.55 (14.60)	0.71 (62.22)
π	5.88 (30.95)	-0.76 (-6.81)	-0.47 (-4.18)	-0.78 (-7.47)	0.58 (4.20)	-1.53 (-12.46)	-0.22 (-3.54)
σ	-0.78 (-91.22)	-0.39 (-13.05)	-0.38 (-12.55)	-0.40 (-26.79)	0.04 (13.35)	-0.45 (-43.01)	-0.36 (-13.64)
α	-0.32 (-7.30)	-0.47 (-2.28)	-0.47 (-2.28)	-0.47 (-10.25)	0.11 (5.66)	-0.60 (-17.50)	-0.33 (-4.84)
q_{RC}	3.67 (3.54)	-3.16 (-0.64)	-3.14 (-0.63)	-3.28 (-3.09)	2.53 (2.90)	-6.56 (-6.48)	-0.11 (-0.08)
q_A	-0.17 (-10.94)	-0.28 (-3.93)	-0.29 (-3.93)	-0.29 (-16.53)	0.04 (8.67)	-0.34 (-26.59)	-0.24 (-8.48)
R^2	0.89 (1)	0.07 (1)	0.07 (1)	0.08 (35)	0.01 (35)	0.06 (35)	0.10 (35)
N	3,000	3,000	3,000	3,000	—	3,000	3,000

Table V
Leverage and Stock Returns

The table reports the results of the following cross-sectional regressions on the level of the quasi-market leverage ratio, QML .

$$QML_{it} = f_0 + f_1 IDR_{t-k,t} + f_2 QML_{t-k} + \epsilon.$$

Independent variables are the implied debt ratio ($IDR_{t-k,t}$) and lagged quasi-market leverage ratio (QML_{t-k}). Coefficients and t -statistics in Panel A are means over 1,000 simulations. Row 1 of Panel B reports Welch's (2004) estimates of the IDR coefficients. Other rows report the mean and the 5th and 95th percentiles of my estimates. One thousand data sets are generated, each containing 75 years of quarterly data for 3,000 firms. For each data set the above regressions are run over the last 35 years of data and then averaged. Standard errors are Fama-MacBeth (1973) with the Newey-West correction.

	h Years			
	1	3	5	10
Panel A				
Constant	0.034 (34.241)	0.088 (41.942)	0.130 (45.289)	0.199 (46.679)
$IDR_{t-k,t}$	1.022 (92.327)	0.886 (75.677)	0.781 (71.356)	0.592 (53.735)
QML_{t-k}	-0.105 (-7.971)	-0.095 (-5.868)	-0.089 (-5.445)	-0.063 (-5.028)
R^2	0.926 (37)	0.802 (35)	0.698 (33)	0.502 (28)
N	3,000	3,000	3,000	3,000
Panel B: IDR Coefficients				
Welch	1.014	0.944	0.869	0.708
This paper	1.022	0.885	0.780	0.591
5%	0.980	0.839	0.735	0.552
95%	1.072	0.935	0.825	0.628

Table IV. Cross-sectional leverage regressions

Table V. Leverage and stock returns

Read: Table IV shows that profitability is positively related to leverage at refinancing points but negatively related to leverage in simulated dynamic cross sections. Table V shows that simulated leverage behaves almost like Welch's implied debt ratio: stock returns mechanically move leverage and firms do not offset the effect in the short run.

Economic message: These are the paper's central empirical exercises. A dynamic trade-off model can generate two patterns often viewed as evidence against trade-off theory: negative leverage-profitability slopes and stock-return-driven leverage.

12.4 Figure 2 and Table VI: implied debt ratio coefficient and mean reversion

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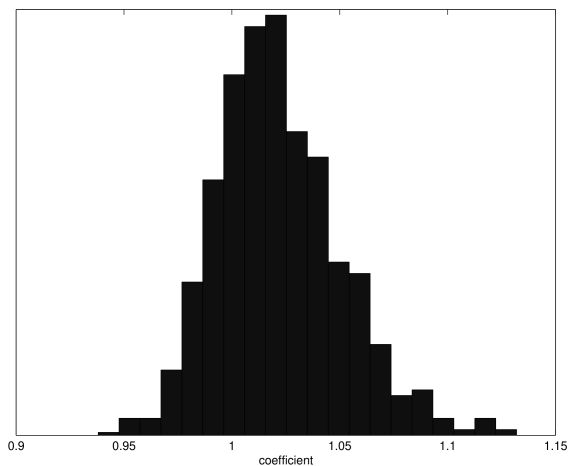


Figure 2. $IDR_{t-1,t}$ coefficient. The figure shows the distribution of the implied debt ratio coefficient in regressions of the quasi-market leverage level against the implied debt ratio and last year's leverage (Welch (2004)).

Figure 2. Distribution of one-year implied debt ratio coefficients

Table VI
Cross-Sectional Regressions for Leverage Changes

The table reports the results of the following Fama-MacBeth (1973) cross-sectional regressions on changes in the quasi-market leverage ratio, $QML_t - QML_{t-1}$:

$$QML_t - QML_{t-1} = h_0 + h_1 TQML_{t-1} + h_2 QML_{t-1} + h_3 X_{t-1} + \epsilon.$$

Independent variables are the target quasi-market leverage ratio ($TQML_{t-1}$), past leverage (QML_{t-1}), implied debt ratio adjustment ($IDR_{t-1,t} - QML_{t-1}$), profitability (π_t), change in profitability ($\Delta\pi_{t-1} = \pi_{t-1} - \pi_{t-2}$), and the cross-term ($(\pi_{t-1} \times (IDR_{t-1,t} - QML_{t-1}))$). One thousand data sets are generated, each containing 75 years of quarterly data for 3,000 firms. For each data set the above regressions are run over the last 35 years of data and then averaged. Coefficients and t -statistics are means over 1,000 simulations.

	(1)	(2)	(3)	(4)
Constant	0.00 (0.54)	0.01 (0.58)	0.03 (12.90)	0.12 (28.40)
$TQML_{t-1}$	0.16 (5.49)	0.16 (-4.98)		
QML_{t-1}	-0.17 (-12.56)	-0.17 (-12.47)		
$IDR_{t-1,t} - QML_{t-1}$			1.02 (85.33)	0.76 (69.77)
π_{t-1}			0.02 (1.38)	0.06 (2.23)
$\Delta\pi_{t-1}$		-1.20 (-7.68)		
$\pi_{t-1} \times (IDR_{t-1,t} - QML_{t-1})$			0.08 (10.78)	0.33 (24.25)
R^2	0.10 (36)	0.13 (36)	0.79 (36)	0.72 (31)
N	3,000	3,000	3,000	3,000

Table VI. Cross-sectional regressions for leverage changes

Read: Figure 2 shows that Welch's one-year IDR coefficient lies well inside the distribution generated by the model. Table VI shows partial-adjustment regressions for leverage changes and a mean-reversion coefficient around -0.17 .

Economic message: Firms can be both active and apparently passive. They adjust infrequently and discretely, producing slow average mean reversion and strong short-run persistence of stock-return-induced leverage changes.

12.5 Table VII and Table VIII: active/passive subsamples and robustness

Table VII
Cross-Sectional Regressions on Subsamples

The table reports the results of cross-sectional regressions on the level of the quasi-market leverage ratio, QML . Independent variables are profitability (π), volatility of cash flows (σ), bankruptcy costs (α), asset sale costs (q_A), and restructuring costs (q_{RC}). The Ref. Point column gives the results obtained by running the regression at the refinancing point. The FF column reports the results of benchmark Fama and French (2002) regression using the Fama–MacBeth (1973) empirical procedure on the whole sample. Column “Active” reports the results of the same regression run on the subsample of firms that restructured over the last year. Column “Passive” reports the results of the same regression run on the subsample of firms that did not restructure over the last year. For each data set the above regressions are run over the last 35 years of data and then averaged. Coefficients and t -statistics are means over 1,000 simulations.

	Ref. Point	FF	Active	Passive
Constant	0.24 (22.29)	0.62 (34.02)	0.60 (143.56)	0.63 (32.77)
π	5.88 (30.95)	-0.78 (-7.47)	-0.03 (-2.48)	-0.84 (-7.89)
σ	-0.78 (-91.22)	-0.40 (-26.79)	-1.09 (-106.80)	-0.37 (-24.77)
α	-0.32 (-7.30)	-0.47 (-10.25)	-0.50 (-15.97)	-0.46 (-9.39)
q_{RC}	3.67 (3.54)	3.28 (-3.09)	5.52 (7.47)	-5.55 (-4.90)
q_A	-0.17 (-10.94)	-0.29 (-16.53)	-0.22 (-18.90)	-0.30 (-16.54)
R^2	0.89 (1)	0.08 (35)	0.77 (35)	0.07 (35)
N	3,000	3,000	346	2,606

Table VII. Cross-sectional regressions on active and passive subsamples

Table VIII
Robustness Tests

The table summarizes the robustness tests. Column (i) reports the difference between average leverage in dynamics and at Ref. Point (See Table III). Column (ii) reports the average value of the profitability coefficient in Fama–French regressions (Table IV). Column (iii) reports the average value of $IDR_{t-1,t}$ (Table V). Column (iv) reports the average value of the mean-reversion coefficient f_2 (Table VI). The tests are as follows: (1) Adjustment cost $q_{RC} = 0.05\%$ for all firms; (2) $q_{RC} = 0.35\%$ for all firms; (3) $q_{RC} = 0.20\%$ for all firms; (4) q_{RC} 's distribution is shifted to the right by 0.15%; (5) Personal tax on interest income $\tau_i = 25\%$; (6) $\tau_i = 40\%$; (7) Interest rate $r = 0.07$; (8) $r = 0.03$; (9) Asset risk premia are 0.09%; (10) Asset risk premia are 0.04%; (11) Volatility of systematic component $\sigma_S = 0.18\%$ (mean of total volatility, σ , is 0.299 with std. dev. of 0.106); (12) $\sigma_S = 0.18$ (σ_j is multiplied by $0.9 \times (\sigma_j - 0.05)$ so that σ has mean and std. dev. similar to the benchmark case); (13) $\sigma_S = 0.05$ (mean of σ is 0.23 with std. dev. of 0.106); (14) $\sigma_S = 0.05(\sigma_j$ is multiplied by $0.93 \times (\sigma_j + 0.043)$ so that σ has mean and std. dev. similar to the benchmark case); (15) $\sigma_S = 0.0$ (σ_j is multiplied by $0.91 \times (\sigma_j + 0.055)$ so that σ has mean and std. dev. similar to the benchmark case); (16) σ has a mean of 0.35 and std. dev. of 0.14 by multiplying original volatility by 1.37; (17) σ has a mean of 0.15 and std. dev. of 0.06 by multiplying original volatility by 0.59; (18) σ is 0.247 for all firms (by taking the mean of σ_j and β); (19) β distribution is wider by 50% (so that σ has a mean of 0.288 and std. dev. of 0.104); (20) β distribution is wider by 50% (σ_j shifts to the right by 0.05 so that σ has mean and std. dev. similar to the benchmark case); (21) All firms have the same firm-specific parameters, which are equal to the mean of the corresponding parameters; (22) $\tau_i = 0$ for all firms; and (23) net payout ratio is 0.035 for all firms. Each test is run for 3,000 firms and 50 economies. In each test, other parameter values and empirical procedures are unchanged.

Test	Description	(i)	(ii)	(iii)	(iv)
1	$q_{RC} = 0.05\%$ for all firms	0.13	-0.62	1.02	-0.16
2	$q_{RC} = 0.35\%$ for all firms	0.10	-0.61	1.04	-0.17
3	$q_{RC} = 0.20\%$ for all firms	0.11	-0.69	1.03	-0.17
4	q_{RC} : distribution is shifted to the right	0.09	-0.53	1.04	-0.17
5	$\tau_i = 25\%$	0.14	-0.60	1.02	-0.22
6	$\tau_i = 40\%$	0.06	-0.72	1.05	-0.12
7	$r = 0.07$	0.08	-0.43	0.99	-0.18
8	$r = 0.03$	0.14	-0.81	1.06	-0.16
9	Asset risk premia are 0.09%	0.08	-0.45	0.99	-0.17
10	Asset risk premia are 0.04%	0.14	-0.70	1.07	-0.17
11	$\sigma_S = 0.18$ (and thus σ is larger)	0.13	-1.07	1.06	-0.17
12	$\sigma_S = 0.18$ (and σ_j is smaller so that σ does not change)	0.11	-0.64	1.07	-0.15
13	$\sigma_S = 0.05$ (and thus σ is smaller)	0.09	-0.29	1.00	-0.17
14	$\sigma_S = 0.05$ (and σ_j is larger so that σ does not change)	0.11	-0.39	1.01	-0.17
15	$\sigma_S = 0$ (and σ_j is larger so that σ does not change)	0.11	-0.33	1.01	-0.17
16	σ has a mean of 0.35	0.09	-0.94	1.03	-0.12
17	σ has a mean of 0.15	0.12	-0.12	1.00	-0.31
18	$\sigma = 0.247$ for all firms	0.09	-1.32	1.02	-0.20
19	β has a wider distribution (so that σ increases)	0.12	-0.79	1.04	-0.17
20	β has a wider distribution (σ_j is changed so that σ is the same)	0.10	-0.48	1.05	-0.16
21	All firm-specific parameters equal to their cross-sectional means	0.10	-0.80	1.00	-0.16
22	No loss of tax shelter, $\tau_i = 0$	0.19	-0.89	1.02	-0.20
23	Net payout ratio is 0.035 for all firms	0.12	-0.66	1.01	-0.17

Table VIII. Robustness tests

Read: Table VII separates active refinancers from passive firms. The active-firm regressions have much higher explanatory power, and the profitability coefficient almost disappears. Table VIII varies costs, tax rates, interest rates, risk premia, systematic risk, volatility, and other assumptions.

Economic message: The misleading negative profitability relation is largely a passive-firm phenomenon. The core results are robust to many parameter changes, especially the Welch-style IDR coefficient and mean-reversion estimates.

13 Short summary

- The paper asks whether standard capital structure tests mean what researchers think they mean.
- The key issue is infrequent adjustment: observed leverage usually differs from refinancing-point target leverage.
- A calibrated dynamic trade-off model is used to simulate panel data for many firms over many years.
- The simulated data reproduce major empirical regularities: negative leverage-profitability relations, stock-return-driven leverage, and slow mean reversion.

- These patterns arise even though firms in the model follow optimal dynamic trade-off policies.
- Therefore, common empirical tests can reject the very model that generated the data.
- The paper argues for dynamic, state-contingent empirical tests that condition on firms' positions in their refinancing cycles.

14 Conclusion

14.1 Core conclusion

Standard tests of capital structure can be misleading because they often compare observed leverage with theoretical predictions at refinancing points. In a dynamic world with adjustment costs, most firms are away from refinancing points, and observed leverage is shaped by past shocks and past financial decisions.

14.2 Incremental contribution revisited

The paper's contribution is not simply another capital structure model. Its contribution is to show how to translate a model into the same kind of data used by empirical researchers, and then to test whether the empirical methods recover the model's true implications.

14.3 Implications for empirical research

- Empirical hypotheses should come from simulated or analytically derived dynamic cross sections.
- Researchers should distinguish active refinancers from passive firms.
- Duration models and refinancing-hazard models may be more informative than unconditional cross-sectional leverage regressions.
- Leverage regressions should account for the beginning and end of refinancing cycles, not only calendar time.

14.4 Economic intuition

The central intuition is that inaction is an action. Firms that do not refinance are not necessarily ignoring target leverage; they may be optimally waiting because adjustment is costly. This inaction changes the empirical relation between leverage and firm characteristics.

14.5 Limitations and future directions

The paper points to two major future directions. First, alternative capital structure theories need fully dynamic quantitative versions. Second, future models should combine dynamic capital structure with real investment and cash policy.

14.6 Takeaway

A negative leverage-profitability relation, slow mean reversion, and the strong role of stock returns in market leverage are not necessarily evidence against dynamic trade-off theory. They may be exactly what optimal dynamic trade-off behavior looks like when observed in a random cross section.